The Very Standard Tricks in Number Theory

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Contents

1 Divisibilities

Divisibility is an important tool in number theory. We need to handle this in many cases, and so it is important to know how to deal with the basic things.

Problem 1.1.

(a) Find all integers n such that $n - 3$ divides $n^2 - 2n$.

(b) Find all integers n such that $4n - 2$ divides $5n + 3$.

(c) Find all integers n such that $2n^2 + 1$ divides $2n^3 + 3n^2 - n + 1$.

Solution. We first go through some divisibility problems involving a single variable. Such problems can always be solved using the methods introduced below.

(a) By the remainder theorem, the remainder when $n^2 - 2n$ is divided by $n - 3$ is $(3)^2 - 2(3) = 3$. Thus, $n - 3 \mid n^2 - 2n$ if and only if $n - 3 \mid 3$. This gives the only possibilities $n - 3 = \pm 1, \pm 3$ (do not miss out the negative solutions), which yields all solutions $n = 0, 2, 4, 6$. □

(b) If we apply the remainder theorem, we need $4n-2$ divides $5\left(\frac{2}{4}\right)+3 = \frac{11}{2}$. This is not the result we want since this is only the remainder obtained by polynomial division. So we may obtain something which is not an integer. (In fact, $\frac{2}{4}$ here is rather different from $\frac{1}{2}$!)

Indeed, noticing that there is a coefficient 4, we first multiply the expression by 4. Since $4n - 2 \mid 5n + 3$, we have

$$4n - 2 \mid 4(5n + 3) = 20n + 12 = 5(4n - 2) + 22.$$

This implies $4n - 2 \mid 22$. Since $4n - 2$ is even, we have $4n - 2 = \pm 2, \pm 22$. This yields $n = -5, 0, 1, 6$. However, the deduction is not reversible in this case (as $4n-2 \mid 20n+12$ does not imply $4n-2 \mid 5n+3$). So we need to check the solutions. Actually, $4n - 2 \mid 5n + 3$ does not hold when $n = 0, 6$. The only solutions are $n = -5, 1$, where $-22 \mid -22$ and $2 \mid 8$. □

(c) As before, we first use long division to obtain

$$2n^3 + 3n^2 - n + 1 = (2n^2 + 1)\left(n + \frac{3}{2}\right) - 2n - \frac{1}{2}.$$

Since we are dealing with integers, we should multiply this by 2. Now, we have

$$2n^2 + 1 \mid 2(2n^3 + 3n^2 - n + 1) = (2n^2 + 1)(2n + 3) - (4n + 1).$$

This implies $2n^2+1 \mid 4n+1$. Here, we use the following fact: $a \mid b$ implies $b = 0$ or $|a| \leqslant |b|$. Note that it is impossible that $4n+1 = 0$.

If $4n+1 > 0$, then $2n^2+1 \leqslant 4n+1$. This implies $n^2-2n \leqslant 0$, and hence $0 \leqslant n \leqslant 2$. The only possibilities are $n = 0, 1, 2$. We check that $1 \mid 1$ when $n = 0$, $3 \nmid 5$ when $n = 1$, and $9 \mid 27$ when $n = 2$.

If $4n+1 < 0$, then $2n^2+1 \leqslant -(4n+1)$. This implies $n^2+2n+1 \leqslant 0$, which means $(n+1)^2 \leqslant 0$. The only possibility is $n = -1$. One checks that $3 \mid 3$ in that case.

Therefore, the only solutions are $n = -1, 0, 2$. $\square$

In general, if we need to solve $P(n) \mid Q(n)$ for some polynomials P and Q, we first apply the division algorithm and find the remainder $R(n)$ when $Q(n)$ is divided by $P(n)$. Multiplying $R(n)$ by a suitable constant, we get a polynomial $S(n)$ with integer coefficients. Then it suffices to solve $P(n) \mid S(n)$, and check solutions at the end. This implies $S(n) = 0$ or $|P(n)| \leqslant |S(n)|$. Clearly, $S(n) = 0$ has finitely many roots. Also, since $\deg P > \deg S$, there are finitely many integers n satisfying $|P(n)| \leqslant |S(n)|$. Thus, we can find some bounds for n and test for all possible values one by one.

Problem 1.2.

(a) Prove that $2n^3+4n^2-11n+22$ and $n^3+2n^2-5n+13$ are relatively prime for any integer n.

(b) Let n be a positive integer. Find the greatest common divisor of $2n^3+7n^2+14n+24$ and n^2+2n+4.

Solution. Finding the greatest common divisor of two integers is another important tool. We can apply the Euclidean algorithm to do so.

(a) Firstly, we have

$$2n^3+4n^2-11n+22 = 2(n^3+2n^2-5n+13) - (n+4).$$

Thus, we have $(2n^3+4n^2-11n+22, n^3+2n^2-5n+13) = (n+4, n^3+2n^2-5n+13)$. Secondly, we have

$$n^3+2n^2-5n+13 = (n^2-2n+3)(n+4)+1.$$

This implies $(n+4, n^3+2n^2-5n+13) = (n+4, 1)$. Since the only positive divisor of 1 is 1, this greatest common divisor must be 1. Therefore, $2n^3+4n^2-11n+22$ and $n^3+2n^2-5n+13$ are relatively prime. $\square$

(b) Similarly, we find that

$$2n^3 + 7n^2 + 14n + 24 = (2n + 3)(n^2 + 2n + 4) + 12.$$

This implies $d = (2n^3 + 7n^2 + 14n + 24, n^2 + 2n + 4) = (12, n^2 + 2n + 4)$. Thus, the greatest common divisor must be a positive divisor of 12. Its exact value depends on n. We need to examine whether n is divisible by 3 and 4.

For divisibility by 3, it suffices to check $n \equiv 0, 1, 2 \pmod 3$. Correspondingly, we have $n^2 + 2n + 4 \equiv 1, 1, 0 \pmod 3$. Therefore, $3 \mid d$ if and only if $n \equiv 2 \pmod 3$.

For divisibility by 4, it suffices to check $n \equiv 0, 1, 2, 3 \pmod 4$. Correspondingly, we have $n^2 + 2n + 4 \equiv 0, 3, 0, 3 \pmod 4$. Therefore, $4 \mid d$ if and only if n is even.

Then we can use the Chinese remainder theorem to combine the constraints. For example, $d = 1$ if $3 \nmid n^2 + 2n + 4$ and $4 \nmid n^2 + 2n + 4$. This means $n \equiv 0, 1 \pmod 3$ and n is odd, i.e. $n \equiv 1, 3 \pmod 6$. Similarly, we find that

$$d = \begin{cases} 1 & \text{if } n \equiv 1, 3 \pmod 6, \\ 3 & \text{if } n \equiv 5 \pmod 6, \\ 4 & \text{if } n \equiv 0, 4 \pmod 6, \\ 12 & \text{if } n \equiv 2 \pmod 6. \end{cases}$$

□

Problem 1.3. Let m and n be positive integers. Prove that $2^m - 1 \mid 2^n - 1$ if and only if $m \mid n$.

Solution. We can transform a divisibility to a congruence. Using congruences is more preferable when the divisor is fixed. We need to simplify $2^n - 1 \pmod{2^m - 1}$.

Let $n = qm + r$ where $0 \leqslant r < m$. Since $2^m \equiv 1 \pmod{2^m - 1}$, we have

$$2^n - 1 = 2^{qm+r} - 1 = (2^m)^q \cdot 2^r - 1 \equiv 1^q \cdot 2^r - 1 = 2^r - 1 \pmod{2^m - 1}.$$

This shows $2^m - 1 \mid 2^n - 1$ if and only if $2^m - 1 \mid 2^r - 1$. Since $0 \leqslant r < m$, we must have $0 \leqslant 2^r - 1 < 2^m - 1$. Therefore, the divisibility holds if and only if $2^r - 1 = 0$, i.e. $r = 0$. This means exactly $m \mid n$. So the equivalence is established. □

Problem 1.4. Find all positive integers a, b, c such that $a \mid b+c$, $b \mid c+a$ and $c \mid a+b$.

Solution. We first introduce two techniques to reduce the number of cases. Firstly, we may assume $\gcd(a, b, c) = 1$ since all the terms have the same degree. Indeed, if $d = \gcd(a, b, c)$, we let $a = da'$, $b = db'$ and $c = dc'$ so that $\gcd(a', b', c') = 1$. Note that $a \mid b + c$ if and only if $a' \mid b' + c'$. Similarly, we have $b' \mid c' + a'$ and $c' \mid a' + b'$. The problem is exactly the same, and we have the extra condition $\gcd(a', b', c') = 1$. Therefore, we can always make this assumption.

Secondly, due to symmetry, we may assume $a \leqslant b \leqslant c$.

- If $a = b = c$, since we need $\gcd(a, b, c) = 1$, the only possibility is $a = b = c = 1$. Clearly, all divisibilities hold.

- If a, b, c are not all equal, we must have $a < c$. Since c is the largest term, we should better consider $c \mid a + b$ first. Indeed, as $2c = c + c > a + b$, we must have $c = a + b$ (recall that $c \mid a + b$ means $a + b = kc$ for some integer k, and clearly k is positive). Thus, the other divisibilities become $a \mid b + a + b$ and $b \mid a + b + a$, which means $a \mid 2b$ and $b \mid 2a$.

  Now, note that $(a, b) = 1$. Otherwise, if $d = (a, b) > 1$, then $d \mid a + b = c$, and so $d \mid \gcd(a, b, c)$, contradicting the assumption. Thus, $a \mid 2b$ implies $a \mid 2$, and similarly $b \mid 2$. It suffices to check $(a, b) = (1, 1), (1, 2), (2, 2)$. The case $a = b = 2$ is rejected since $(a, b) = 1$.

  When $a = b = 1$, we have $c = 2$. We check that $1 \mid 1 + 2$ and $2 \mid 1 + 1$.

  When $a = 1$ and $b = 2$, we have $c = 3$. We check that $1 \mid 2+3$, $2 \mid 3+1$ and $3 \mid 1+2$.

Lastly, we can remove the constraint $\gcd(a, b, c) = 1$ by multiplying all variables by the same constant. Thus, the solutions are $(a, b, c) = (d, d, d), (d, d, 2d), (d, 2d, 3d)$ and their permutations, where $d \in \mathbb{N}$. $\square$

2 Diophantine Equations

One common type of number theoretic problems is to solve Diophantine equations. This means to solve equations where the variables are integers or natural numbers. Solving Diophantine equations is quite different from solving equations in real numbers, since we can use divisibility to deduce more results. Take a simple example $xy = 10$. If x, y are real numbers, then there are infinitely many solutions given by $(x, y) = \left(t, \frac{10}{t}\right)$ for $t \neq 0$. If x, y are positive integers, then we only have finitely many solutions $(x, y) = (1, 10), (2, 5), (5, 2), (10, 1)$.

2.1 Factorization

In view of the above example, one trick in solving Diophantine equations is to factorize the expressions and compare the divisors on both sides.

Problem 2.1.

(a) Solve the equation $ab - 5a - 6b + 18 = 0$ in integers.

(b) Solve the equation $2ab - a + 3b - 4 = 0$ in integers.

Solution.

(a) We try to factorize the expression $ab - 5a - 6b + 18$. By noting $ab - 5a = a(b - 5)$, we take out the factor $b - 5$. Then we rewrite $-6b + 18 = -6(b - 5) - 12$. This gives $ab - 5a - 6b + 18 = (a - 6)(b - 5) - 12$. Although this is not really a factorization, the result is good enough, since we can put the constant on the other side to get

$$(a - 6)(b - 5) = 12.$$

There is a finite number of ways to express 12 as a product of 2 integers. Since we are working on integers, we have

$$(a - 6, b - 5) = (\pm 1, \pm 12), (\pm 2, \pm 6), (\pm 3, \pm 4), (\pm 4, \pm 3), (\pm 6, \pm 2), (\pm 12, \pm 1).$$

These give all the 12 solutions

$$(a, b) = (7, 17), (8, 11), (9, 9), (10, 8), (12, 7), (18, 6), (5, -7), (4, -1), (3, 1), (2, 2), (0, 3), (-6, 4).$$

Even if we are solving the equation in positive integers, we still need to consider the negative factorizations of 12. For example, $(-3) \times (-4) = 12$ corresponds to the positive integer solution $(a, b) = (3, 1)$. $\square$

(b) Using the same consideration, we rewrite $2ab - a + 3b - 4 = \left(a + \frac{3}{2}\right)(2b-1) - \frac{5}{2}$. This is not appropriate since we are working on integers. Just like the situation in divisibility problems, we should multiply both sides by 2. Thus, we need to solve

$$(2a+3)(2b-1) = 5.$$

This implies $(2a+3, 2b-1) = (\pm1, \pm5), (\pm5, \pm1)$, and so there are 4 solutions

$$(a,b) = (-1,3), (1,1), (-2,-2), (-4,0).$$

In general, the same method can be used to solve equations of the form

$$\alpha ab + \beta a + \gamma b + \delta = 0.$$

□

Problem 2.2.

(a) Solve the equation $m^2 - n^2 = 12$ in integers.

(b) Solve the equation $3m^2 + 5mn - 2n^2 - 5m + 4n = 0$ in integers.

Solution.

(a) Clearly, the left-hand side can be factorized to give $(m-n)(m+n) = 12$. By noting that $m-n$ and $m+n$ have the same parity, we only need to consider

$$(m-n, m+n) = (\pm2, \pm6), (\pm6, \pm2).$$

Therefore, the solutions are $(m,n) = (\pm4, \pm2)$ (where the two signs are independent). □

(b) Indeed, the equation is the same as

$$(3m - n + 1)(m + 2n - 2) = -2.$$

If one cannot spot the factorization immediately, one can regard it as an equation in real numbers, and use the quadratic formula to solve it. For example, rewrite this as an equation in m as follows:

$$3m^2 + (5n-5)m + (-2n^2 + 4n) = 0.$$

By the quadratic formula, we obtain

$$m = \frac{(5-5n) \pm \sqrt{(5n-5)^2 - 4(3)(-2n^2+4n)}}{2(3)} = \frac{(5-5n) \pm \sqrt{49n^2 - 98n + 25}}{6}.$$

Since m is an integer, the thing inside the radical, i.e. $49n^2 - 98n + 25$, is a perfect square. Let $49n^2 - 98n + 25 = k^2$ for some nonnegative integer k. This implies $(7n-7)^2 - 24 = k^2$. By rearranging the terms, this is the same as

$$(7n - 7 - k)(7n - 7 + k) = (7n-7)^2 - k^2 = 24.$$

As $7n-7-k$ and $7n-7+k$ have the same parity, this yields

$$(7n-7-k, 7n-7+k) = (\pm 2, \pm 12), (\pm 4, \pm 6), (\pm 6, \pm 4), (\pm 12, \pm 2).$$

The only solutions for which $n \in \mathbb{Z}$ and $k \geqslant 0$ are $(n,k) = (0,5), (2,5)$. This corresponds to $(m,n) = (0,0), (0,2)$. Both of them are solutions by checking. □

Problem 2.3.

(a) Solve the equation $4m^2 - n^3 = 1$ in integers.

(b) Solve the equation $n^3 + n = m^7$ in integers.

Solution. Here we are going to use the following fact: if $AB = C^k$ for some positive integers A, B, C, k where $(A, B) = 1$, then each of A and B is the kth power of a positive integer. This can be easily explained by considering the prime factorization of both sides, since the exponent of each prime must be a multiple of k, and this exponent must come entirely from A or B.

We can generalize this to integer solutions. If k is odd, then the same result holds for negative integers A, B, C. If k is even and A, B, C can be negative, then the conclusion is that $\pm A$ and $\pm B$ are the kth powers of some integers. In both cases, the special case $C = 0$ (and $A = 0$ or $B = 0$) should be handled separately.

(a) There are 2 ways to factorize some of the terms, either factorizing $4m^2 - 1$ or $n^3 + 1$. We choose the former in this solution. Thus, we rewrite the equation as

$$(2m-1)(2m+1) = 4m^2 - 1 = n^3.$$

Now, note that $(2m-1, 2m+1) = (2m-1, 2) = 1$. Using the above fact, we know that each of $2m-1$ and $2m+1$ is a cube (also because $n \neq 0$). As they differ by 2, the only possibility is $2m-1 = -1$ and $2m+1 = 1$, i.e. $m = 0$. Correspondingly, we get the only solution $(m,n) = (0,-1)$. □

(b) This seems to be hard as it involves some high powers. But then it becomes simple by using the above fact. First we rewrite the equation as

$$n(n^2+1) = m^7.$$

Note that $(n, n^2+1) = (n, 1) = 1$. Thus, we know that both n and n^2+1 are 7th powers unless one of them is 0. The exceptional case is easy since we must have $n = 0$, and hence $m = 0$. Suppose $n \neq 0$. Let $n = a^7$ and $n^2 + 1 = b^7$. This implies $b^7 - a^{14} = 1$. Thus, b^7 and $a^{14} = (a^2)^7$ are two 7th powers with difference 1. The only possibilities are $(b^7, a^{14}) = (1,0), (0,-1)$. As $a^{14} \geqslant 0$, the latter case is rejected. The former case is also rejected since $n \neq 0$. (Alternatively, the equation $b^7 - a^{14} = 1$ only has the trivial solution by the Fermat last theorem!)

Therefore, the only solution is $(m,n) = (0,0)$. □

2.2 Congruences

Considering the divisibility by some small numbers like $2, 3, 4, 8$ is another useful tool in solving Diophantine equations. This allows us to conclude the equation has no solution, or some variables must satisfy certain constraints if a solution exists.

Problem 2.4.

(a) Solve the equation $a^2 + 3b^2 = 2345$ in integers.

(b) Solve the equation $a^3 + b^3 + c^3 = 2345$ in integers.

Solution. We shall show that both equations have no solution. The trick is to find some n such that both sides modulo n cannot be congruent to each other. This clearly implies a solution cannot exist. However, the converse does not hold. That means even if the equation is solvable modulo n for every positive integer n, we cannot conclude the equation is solvable in integers.

(a) For square numbers or quadratic polynomials, it is common to consider modulo $3, 4, 5, 8$. For example, by considering modulo 3, we find that

$$a^2 \equiv 2 \pmod 3.$$

But we know that a square number must be congruent to 0 or 1 modulo 3. This is a contradiction. Therefore, the Diophantine equation has no solution. □

(b) For cubes, it is common to consider modulo 7 or 9. Indeed, we must have $k^3 \equiv -1, 0, 1 \pmod 7$ and $k^3 \equiv -1, 0, 1 \pmod 9$ (for example, we can compute $0^3, 1^3, \ldots, 8^3 \pmod 9$). By considering modulo 9, we find that

$$a^3 + b^3 + c^3 \equiv 5 \pmod 9.$$

But since each cube is congruent to $-1, 0, 1$ modulo 9, we can only have

$$a^3 + b^3 + c^3 \equiv -3, -2, -1, 0, 1, 2, 3 \pmod 9.$$

This is a contradiction. Therefore, the Diophantine equation has no solution. □

Problem 2.5. Solve the equation $a^3 - b^3 = 123456$ in integers.

Solution. Using the factorization $a^3 - b^3 = (a-b)(a^2+ab+b^2)$, we may try to factorize 123456 into 2 factors and solve the 2 equations in 2 unknowns. However, there are a lot of cases to consider, each of which involves a quadratic equation. One simpler way is to consider congruences and reduce some (or even all) cases.

By considering modulo 3, we get $a - b \equiv 0 \pmod 3$ since $a^3 \equiv a \pmod 3$ for any a (one may check directly by putting $a \equiv 0, 1, 2 \pmod 3$ or use the Fermat little theorem). Let $a = b + 3k$ for some integer k. Then we have

$$a^3 - b^3 = (b + 3k)^3 - b^3 = b^3 + 9b^2k + 27bk^2 + 27k^3 - b^3 = 9(b^2k + 3bk^2 + 3k^3).$$

This shows $9 \mid a^3 - b^3$. But then $123456 \equiv 1+2+3+4+5+6 \equiv 3 \not\equiv 0 \pmod 9$. Thus, the equation has no integer solution. $\square$

Problem 2.6. Find the minimum positive integer k such that there exist prime numbers $p_1, p_2, \dots, p_k$ (not necessarily distinct) satisfying $p_1^2 + p_2^2 + \cdots + p_k^2 = 2012^2$.

Solution. The key idea is that all primes except 2 are odd, and the square of an odd number must be congruent to 1 modulo 8. Therefore, it is natural to consider modulo 8.

As explained, we have $p_j^2 \equiv 1, 4 \pmod 8$, and it is 4 if and only if $p_j = 2$. Since

$$p_1^2 + p_2^2 + \cdots + p_k^2 \equiv 0 \pmod 8,$$

we must have $k = 2$ (corresponding to the case $4 + 4 \equiv 0 \pmod 8$) or $k \geqslant 5$ (corresponding to the minimal case $4+1+1+1+1 \equiv 0 \pmod 8$). The former case is rejected since the only possibility is $2^2 + 2^2$, which is less than 2012^2.

The case $k = 5$ can be attained. The construction does not require much knowledge in number theory except the understanding of prime numbers. One way to construct the numbers is to start from some large primes. For example, note that 2011 is a prime. So we consider $p_5 = 2011$. This gives

$$p_1^2 + p_2^2 + p_3^2 + p_4^2 = 2012^2 - 2011^2 = (2012 - 2011)(2012 + 2011) = 4023.$$

Note that exactly one of the primes is 2 since the sum is odd. By considering modulo 3 (as $p^2 \equiv 1 \pmod 3$ except $p = 3$), we know that one of the primes is 3. It is not hard to work out one example $2^2 + 3^2 + 17^2 + 61^2 + 2011^2 = 2012^2$. $\square$

Problem 2.7. Solve the equation $2^x - 3^y = 1$ in positive integers.

Solution. Clearly, $(x, y) = (2, 1)$ is a solution. So there is no hope to show that the equation has no solution in integers by considering any congruences. However, we can still use this trick.

We would like to show that there is no solution when $x \geqslant 3$. In view of the difference between $x \geqslant 3$ and $x \leqslant 2$, we should consider modulo 8. Indeed, if $x \geqslant 3$, then we have $8 \mid 2^x$ and hence $-3^y \equiv 1 \pmod 8$. The powers of 3 modulo 8 are $3, 1, 3, 1, \dots$, which repeat every 2 powers. In particular, we see that 3^y can never be congruent to -1 modulo 8. (Indeed, for any positive integers a and b, the sequence $a, a^2, a^3, \dots$ modulo b is always periodic. We can list out all possible terms and deduce some properties of the exponents.) This means $x \leqslant 2$.

When $x = 1$, $y = 0$ is not positive. When $x = 2$, $y = 1$ is the only solution. $\square$

2.3 Bounding Variables

Another standard method is to obtain some bounds (usually upper bounds) for the variables through analyzing the equations. Once this is done, usually there is only a finite number of cases left, which can be handled by exhaustion. This method is used when there is a term which dominates the other terms when the variables are large.

Problem 2.8. Solve the equation $abc = 2a + b + c$ in positive integers.

Solution. This seems to be a generalization of problem 2.1. But in most cases, we cannot use the factorization method to obtain $(a - \alpha)(b - \beta)(c - \gamma) = \delta$. Therefore, we shall introduce a more general method to handle such equations. Observe that abc dominates the expression if a, b, c are large.

Rewrite the equation as $a(bc - 2) = b + c$. Since the right-hand side is positive, we must have $bc > 2$. Thus, the equation implies

$$bc - 2 \leqslant a(bc - 2) = b + c.$$

This yields $(b - 1)(c - 1) \leqslant 3$. Without loss of generality, assume $b \geqslant c$.

- If $c = 1$, the equation becomes $ab = 2a + b + 1$, which gives $(a - 1)(b - 2) = 3$. The solutions are $(a, b) = (2, 5), (4, 3)$.

- If $c = 2$, we have $2b > 2$ and $b - 1 \leqslant 3$. This means $1 < b \leqslant 4$, and hence $b = 2, 3, 4$. It is easy to find out the solutions $(a, b) = (2, 2), (1, 4)$.

- If $c \geqslant 3$, we have $(b - 1)(c - 1) \geqslant (3 - 1)(3 - 1) = 4$, contradiction.

Therefore, all the solutions are $(a, b, c) = (2, 5, 1), (2, 1, 5), (4, 3, 1), (4, 1, 3), (2, 2, 2), (1, 4, 2), (1, 2, 4)$. □

Problem 2.9. Solve the equation $n^4 + 2n^3 + 5n^2 + 3n + 12 = m^2$ in positive integers.

Solution. Since we want $n^4 + 2n^3 + 5n^2 + 3n + 12$ to be a perfect square, we try to use the method of completing square to get

$$n^4 + 2n^3 + 5n^2 + 3n + 12 = (n^2 + n + 2)^2 - n + 8.$$

In some sense, m should be close to $n^2 + n + 2$. We can use bounds to make this precise.

Indeed, note that $n^4 + 2n^3 + 5n^2 + 3n + 12 > n^4 + 2n^3 + 3n^2 + 2n + 1 = (n^2 + n + 1)^2$. This yields $m > n^2 + n + 1$.

If $m = n^2 + n + 2$, then we need $-n + 8 = 0$, i.e. $n = 8$. This gives $m = 74$.

If $m \geqslant n^2 + n + 3$, then we have

$$n^4 + 2n^3 + 5n^2 + 3n + 12 \geqslant (n^2 + n + 3)^2 = n^4 + 2n^3 + 7n^2 + 6n + 9.$$

This implies $2n^2 + 3n \leqslant 3$. But this is impossible since $2n^2 + 3n \geqslant 2 + 3 = 5$.

So the only solution is $(m, n) = (74, 8)$. $\square$

3 Construction and Existence Problems

Sometimes we may need to construct some integers with certain properties. There are many different tricks which may be useful, depending on the problems.

Problem 3.1. For each positive integer k, prove that there exist k positive integers (not necessarily distinct) whose sum and product are equal.

Solution. In general, the product of some positive integers should be larger than their sum, unless most numbers are small (like 1 or 2). So we might consider the cases where most numbers are 1 first. Let $a_1, a_2, \ldots, a_k$ be the numbers. If $k-1$ of them are 1, then $k-1+a_k = a_k$ is only solvable when $k=1$, in which case a_1 can be any positive integer.

For $k \geqslant 2$, if $k-2$ of the numbers are 1, we need $k-2+a_{k-1}+a_k = a_{k-1}a_k$, which is the same as $(a_{k-1}-1)(a_k-1) = k-1$. One simple solution is $a_{k-1} = 2$ and $a_k = k$. In order words, we may simply take $1, 1, \ldots, 1, 2, k$ for $k \geqslant 2$. □

Problem 3.2. For each positive integer k, prove that there exist k consecutive composite numbers.

Solution. One standard way is to consider the numbers

$$M+2, M+3, \ldots, M+(k+1).$$

If M is a positive multiple of $2, 3, \ldots, k+1$ (say, $M = (k+1)!$ or the least common multiple of these numbers), then the first term is a multiple of 2, the second term is a multiple of 3, …, and the last term is a multiple of $k+1$. Since all of them are larger than $k+1$, these k consecutive numbers must be composite.

Another common way is to apply the Chinese remainder theorem. Indeed, we can pick distinct primes $p_1, p_2, \ldots, p_k, q_1, q_2, \ldots, q_k$ and try to find the numbers

$$n+1, n+2, \ldots, n+k$$

for which $p_j q_j \mid n+j$ for each $j = 1, 2, \ldots, k$. If this can be done, then these numbers are not primes. Such conditions can be rewritten as

$$n \equiv -j \pmod{p_j q_j}$$

for each $j = 1, 2, \ldots, k$. As the primes are all distinct, they are relatively prime. By the Chinese remainder theorem, such a positive integer n always exists and hence we are done. This method has the advantage that we can control the prime divisors of the numbers. □

Problem 3.3. For each positive integer k, prove that there exists an integer m such that $2^k \mid m^2 + 7$.

Solution. This time we use induction to prove the existence of such an integer m. When $k = 1, 2, 3$, we can simply take $m = 1$ since $8 \mid 1^2 + 7$.

Let $k \geqslant 3$. Assume $2^k \mid a^2 + 7$ for some integer a. We want to find m such that $2^{k+1} \mid m^2 + 7$. Note that this implies $2^k \mid m^2 + 7$. Therefore, it is natural to consider $m \equiv a \pmod{2^k}$. Let $m = a + 2^k b$. Then we have

$$m^2 + 7 = a^2 + 2^{k+1}ab + 2^{2k}b^2 + 7 \equiv a^2 + 7 \pmod{2^{k+1}}.$$

However, this needs not be congruent to 0 because $a^2 + 7$ is fixed, but we can only choose the value of b. In view of this, we can reduce the power to allow one more term to survive under modulo 2^{k+1}.

Thus, rather than considering $m = a + 2^k b$, we let $m = a + 2^{k-1}c$. Then we have

$$m^2 + 7 = a^2 + 2^k ac + 2^{2k-2}c^2 + 7 \equiv 2^k ac + a^2 + 7 \pmod{2^{k+1}}.$$

This time we can choose a suitable c so that this is congruent to 0. Indeed, since $2^k \mid a^2 + 7$, we can define $a^2 + 7 = 2^k d$ for some integer d. This yields

$$m^2 + 7 \equiv 2^k ac + 2^k d = 2^k(ac + d) \pmod{2^{k+1}}.$$

Note that a must be odd since $a^2 + 7$ is even. If d is odd, we choose $c = 1$ (hence $m = a + 2^{k-1}$) so that $2^k(a + d)$ is divisible by 2^{k+1}. If d is even, we choose $c = 0$ (hence $m = a$) so that $2^k d$ is divisible by 2^{k+1}. Therefore, we can always find an integer m such that $2^{k+1} \mid m^2 + 7$. By induction, we are done. $\square$

Problem 3.4. Prove that there exists a positive integer n such that $2^n - 1$ has at least 1000 prime divisors.

Solution. We first examine when $2^n - 1$ is divisible by an odd prime p. A simple sufficient condition is $p - 1 \mid n$ by the Fermat little theorem. This provides a crucial step to solve this problem.

Indeed, let $p_1, p_2, \ldots, p_{1000}$ be distinct prime numbers, and let n be the least common multiple of $p_1 - 1, p_2 - 1, \ldots, p_{1000} - 1$. Then we have $p_j - 1 \mid n$ for each j, and hence $2^n \equiv 1 \pmod{p_j}$ by the Fermat little theorem. This implies $2^n \equiv 1 \pmod{p_1 p_2 \cdots p_{1000}}$, and hence $2^n - 1$ has at least 1000 prime divisors. $\square$

Problem 3.5. For each prime number p, prove that there exists a positive integer n such that $p \mid 2^n - n$.

Solution. This is another application of the Fermat little theorem. This theorem is useful when handling some exponents.

Firstly, for any odd prime p, we know that $2^n \equiv 1 \pmod{p}$ if $p - 1 \mid n$ by the Fermat little theorem. However, we need $p \mid 2^n - n$ instead of $p \mid 2^n - 1$. This can be made true if we have $p \mid n - 1$, since $2^n - 1 = (2^n - n) + (n - 1)$. In other words, we try to find a positive integer n for which $p - 1 \mid n$ and $p \mid n - 1$. Equivalently, we need

$$\begin{cases} n \equiv 0 \pmod{p-1}, \\ n \equiv 1 \pmod{p}. \end{cases}$$

Since $p - 1$ and p are relatively prime, this can always be done by the Chinese remainder theorem. Indeed, we may simply take $n = (p - 1)^2$. (There is no need to find the exact expression for n as the existence is guaranteed by the Chinese remainder theorem.)

Lastly, for $p = 2$, we simply take n to be any positive even number, say $n = 2$. $\square$